

CHAPTER

22 Oscilloscope and Other Measuring Instruments

22.1 INTRODUCTION

One of the basic functions of electronic circuits is the generation and manipulation of electronic waveshapes. These electronic signals may represent audio information, computer data, television signals, timing signals (as used in radar), and so on. The common meters used in electronic measurement are the multimeter—analog or digital, to enable measuring dc or ac voltages, currents, or impedances. Most meters provide ac measurements that are correct for nondistorted sinusoidal signals only. The oscilloscope, on the other hand, displays the exact waveform, and the viewer can decide what to make of the various readings observed.

The cathode ray oscilloscope (CRO) provides a visual presentation of any waveform applied to the input terminals. A cathode ray tube (CRT), much like a television tube, provides the visual display showing the form of the signal applied as a waveform on the front screen. An electron beam is deflected as it sweeps across the tube face, leaving a display of the signal applied to input terminals.

While multimeters provide numeric information about an applied signal, the oscilloscope allows the actual form of the waveform to be displayed. A wide range of oscilloscopes is available, some suited to measure signals below a specified frequency, others to provide measuring signals of the shortest time span. A CRO may be built to operate from a few hertz up to hundreds of megahertz; CROs may also be used to measure time spans from fractions of a nanosecond (10^{-9}) to many seconds.

22.2 CATHODE RAY TUBE — THEORY AND CONSTRUCTION

The CRT is the "heart" of the CRO, providing visual display of an input signal's waveform. A CRT contains four basic parts:

1. An electron gun to produce a stream of electrons.
2. Focusing and accelerating elements to produce a well-defined beam of electrons.
3. Horizontal and vertical deflecting plates to control the path of the electron beam.
4. An evacuated glass envelope with a phosphorescent screen, which glows visibly when struck by the electron beam.

Figure 22.1 shows the basic construction of a CRT. We will first consider the device's basic operation. A cathode (K) containing an oxide coating is heated indirectly

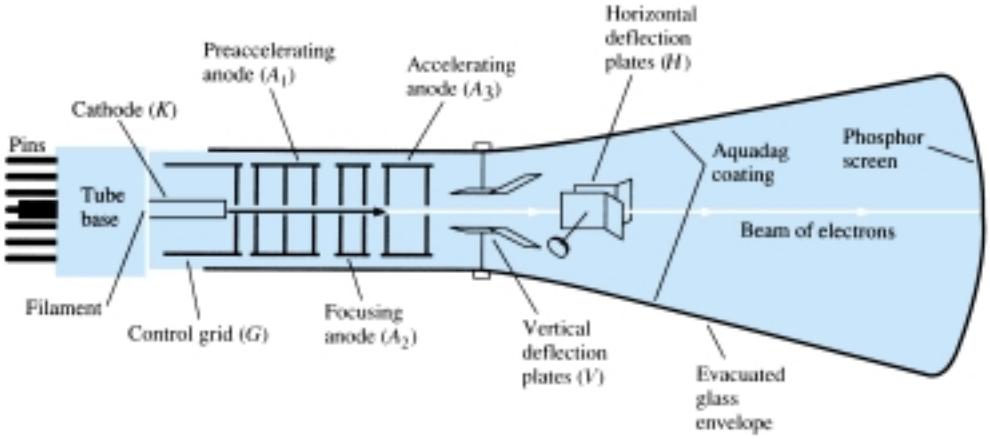


Figure 22.1 Cathode ray tube: basic construction.

by a filament, resulting in the release of electrons from the cathode surface. A control grid (G) provides for control of the number of electrons passing farther into the tube. A voltage on the control grid determines how many of the electrons freed by heating are allowed to continue moving toward the face of the tube. After the electrons pass the control grid, they are focused into a tight beam and accelerated to a higher velocity by the focusing and accelerating anodes. The parts discussed so far comprise the electron gun of the CRT.

The high-velocity, well-defined electron beam then passes through two sets of deflection plates. The first set of plates is oriented to deflect the electron beam vertically, up or down. The direction of the vertical deflection is determined by the voltage polarity applied to the deflecting plates. The amount of deflection is set by the magnitude of the applied voltage. The beam is also deflected horizontally (left or right) by a voltage applied to the horizontal deflecting plates. The deflected beam is then further accelerated by very high voltages applied to the tube, with the beam finally striking a phosphorescent material on the inside face of the tube. This phosphor glows when struck by the energetic electrons—the visible glow seen at the front of the tube by the person using the scope.

The CRT is a self-contained unit with leads brought out through a base to pins. Various types of CRTs are manufactured in a variety of sizes, with different phosphor materials and deflection electrode placement. We can now consider how the CRT is used in an oscilloscope.

22.3 CATHODE RAY OSCILLOSCOPE OPERATION

For operation as an oscilloscope, the electron beam is deflected horizontally by a sweep voltage and vertically by the voltage to be measured. While the electron beam is moved across the face of the CRT by the horizontal sweep signal, the input signal deflects the beam vertically, resulting in a display of the input signal waveform. One sweep of the beam across the face of the tube, followed by a "blank" period during which the beam is turned off while being returned to the starting point across the tube face, constitutes one sweep of the beam.

A steady display is obtained when the beam repeatedly sweeps across the tube with exactly the same image each sweep. This requires a synchronization, starting the sweep at the same point in a repetitive waveform cycle. If the signal is properly synchronized, the display will be stationary. In the absence of sync, the picture will appear to drift or move horizontally across the screen.

Basic Parts of a CRO

The basic parts of a CRO are shown in Fig. 22.2. We will first consider the CRO's operation for this simplified block diagram. To obtain a noticeable beam deflection from a centimeter to a few centimeters, the usual voltage applied to the deflection plates must be on the order of tens to hundreds of volts. Since the signals measured using a CRO are typically only a few volts, or even a few millivolts, amplifier circuits are needed to increase the input signal to the voltage levels required to operate the tube. There are amplifier sections for both the vertical and the horizontal deflection of the beam. To adjust the level of a signal, each input goes through an attenuator circuit, which can adjust the amplitude of the display.

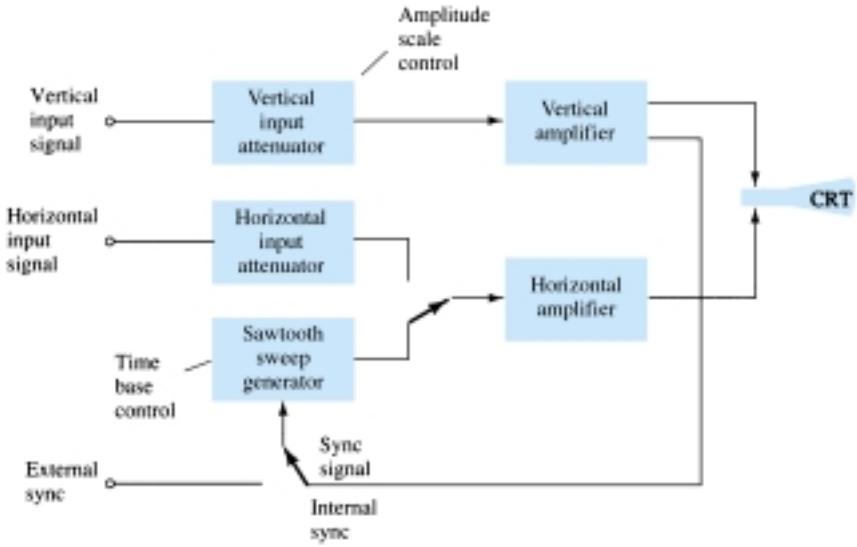


Figure 22.2 Cathode ray oscilloscope: general block diagram.

22.4 VOLTAGE SWEEP OPERATION

When the vertical input is 0 V, the electron beam may be positioned at the vertical center of the screen. If 0 V is also applied to the horizontal input, the beam is then at the center of the CRT face and remains a stationary dot. The vertical and horizontal positioning controls allow moving the dot anywhere on the tube face. Any dc voltage applied to an input will result in shifting the dot. Figure 22.3 shows a CRT face with a centered dot and with a dot moved by a positive horizontal voltage (to the right) and a negative vertical input voltage (down from center).

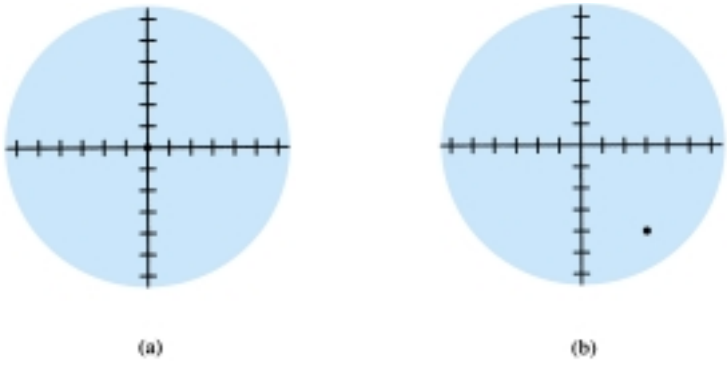


Figure 22.3 Dot on CRT screen due to stationary electron beam: (a) centered dot due to stationary electron beam; (b) off-center stationary dot.

Horizontal Sweep Signal

To view a signal on the CRT face, it is necessary to deflect the beam across the CRT with a horizontal sweep signal so that any variation of the vertical signal can be observed. Figure 22.4 shows the resulting straight-line display for a positive voltage applied to the vertical input using a linear (sawtooth) sweep signal applied to the horizontal channel. With the electron beam held at a constant vertical distance, the horizontal voltage, going from negative to zero to positive voltage, causes the beam to move from the left side of the tube, to the center, to the right side. The resulting display is a straight line above the vertical center with the dc voltage properly displayed as a straight line.

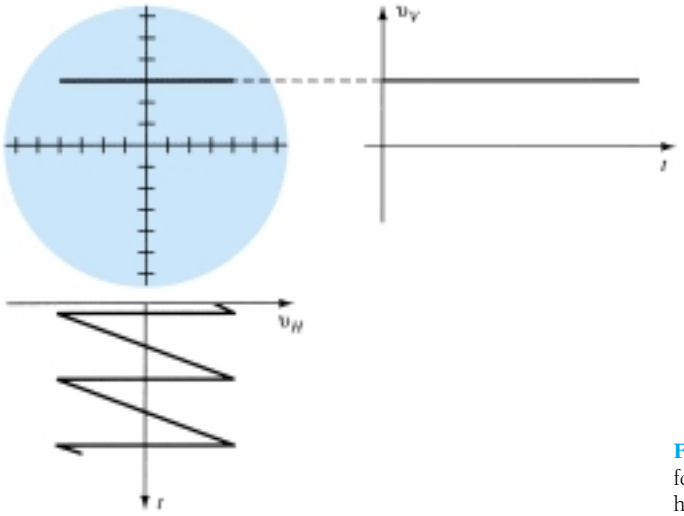


Figure 22.4 Scope display for dc vertical signal and linear horizontal sweep signal.

The sweep voltage is shown to be a continuous waveform, not just a single sweep. This is necessary if a long-term display is to be seen. A single sweep across the tube face would quickly fade out. By repeating the sweep, the display is generated over and over, and if enough sweeps are generated per second, the display appears present continuously. If the sweep rate is slowed down (as set by the time-scale controls of the scope), the actual travel of the beam across the tube face can be observed.

Applying only a sinusoidal signal to the vertical inputs (no horizontal sweep) results in a vertical straight line as shown in Fig. 22.5. If the sweep speed (frequency

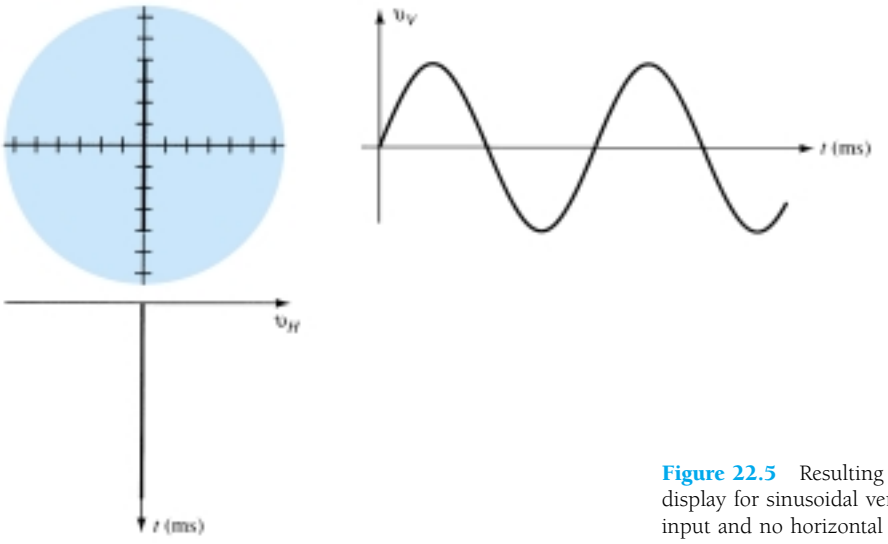


Figure 22.5 Resulting scope display for sinusoidal vertical input and no horizontal input.

of the sinusoidal signal) is reduced, it is possible to see the electron beam moving up and down along a straight-line path.

Use of Linear Sawtooth Sweep to Display Vertical Input

To view a sinusoidal signal, it is necessary to use a sweep signal on the horizontal channel so that the signal applied to the vertical channel can be seen on the tube face. Figure 22.6 shows the resulting CRO display from a horizontal linear sweep and a sinusoidal input to the vertical channel. For one cycle of the input signal to appear as shown in Fig. 22.6a, it is necessary that the signal and linear sweep frequencies be synchronized. If there is any difference, the display will appear to move (not be synchronized) unless the sweep frequency is some multiple of the sinusoidal frequency. Lowering the sweep frequency allows more cycles of the sinusoidal signal to be displayed, whereas increasing the sweep frequency results in less of the sinusoidal vertical input to be displayed, thereby appearing as a magnification of a part of the input signal.

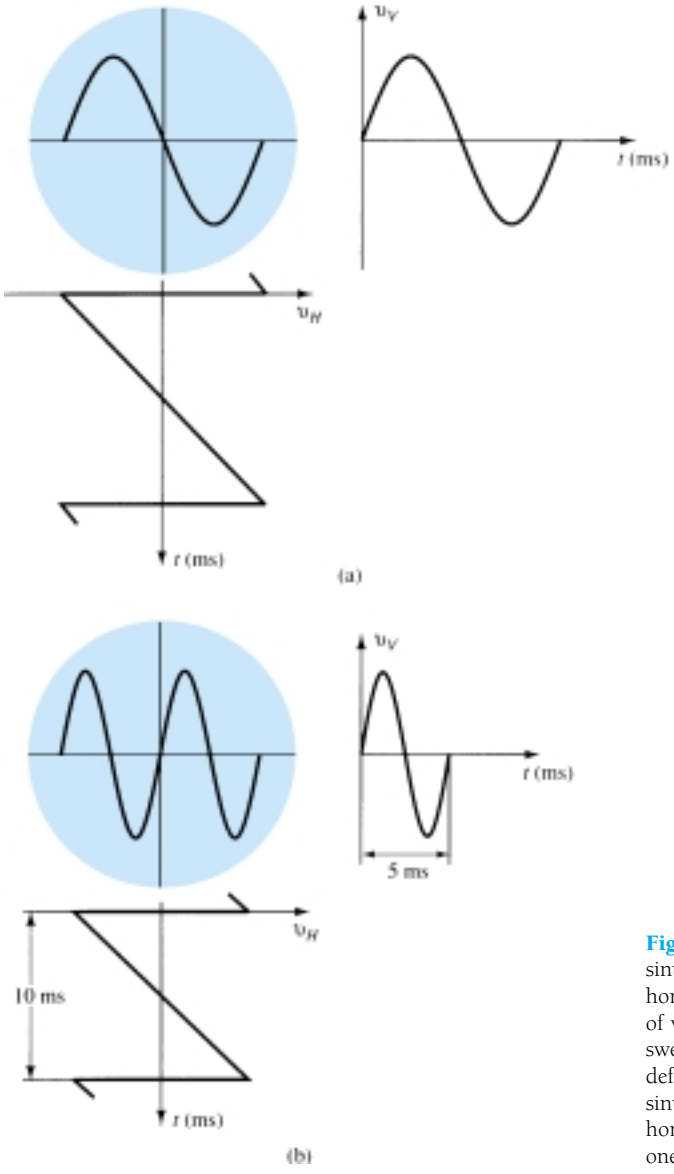


Figure 22.6 Display of sinusoidal vertical input and horizontal sweep input: (a) display of vertical input signal using linear sweep signal for horizontal deflection; (b) scope display for a sinusoidal vertical input and a horizontal sweep speed equal to one-half that of the vertical signal.

EXAMPLE 22.1

Determine how many cycles of a 2-kHz sinusoidal signal are viewed if the sweep frequency is:
(a) 2 kHz.
(b) 4 kHz.
(c) 1 kHz.

Solution

(a) When the two signals have the same frequency, a full cycle will be seen.
(b) When the sweep frequency is increased to 4 kHz, a half-cycle will be seen.
(c) When the sweep frequency is reduced to 1 kHz, two cycles will be seen.

Figure 22.7 shows a pulse-type waveform applied as vertical input with a horizontal sweep, resulting in a scope display of the pulse signal. The numbering at each waveform permits following the display for variation of input and sweep voltage during one cycle.

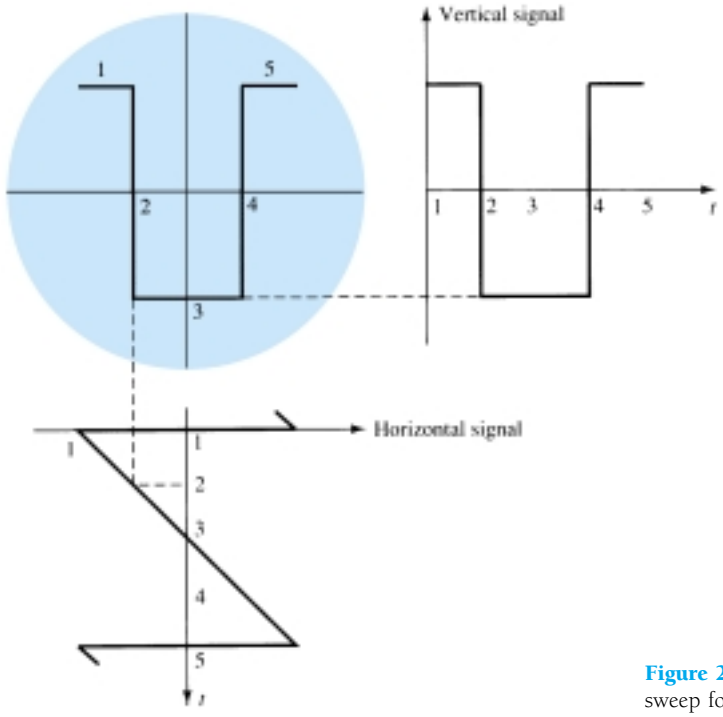


Figure 22.7 Use of the linear sweep for a pulse-type waveform.

22.5 SYNCHRONIZATION AND TRIGGERING

A CRO display can be adjusted by setting the sweep speed (frequency) to display either one cycle, a number of cycles, or part of a cycle. This is a very valuable feature of the CRO. Figure 22.8 shows the display resulting for a few cycles of the sweep signal. Each time the horizontal sawtooth sweep voltage goes through a linear sweep cycle (from maximum negative to zero to maximum positive), the electron beam is caused to move horizontally across the tube face, from left to center to right. The sawtooth voltage then drops quickly back to the negative starting voltage, with the beam back to the left side. During the time the sweep voltage goes quickly

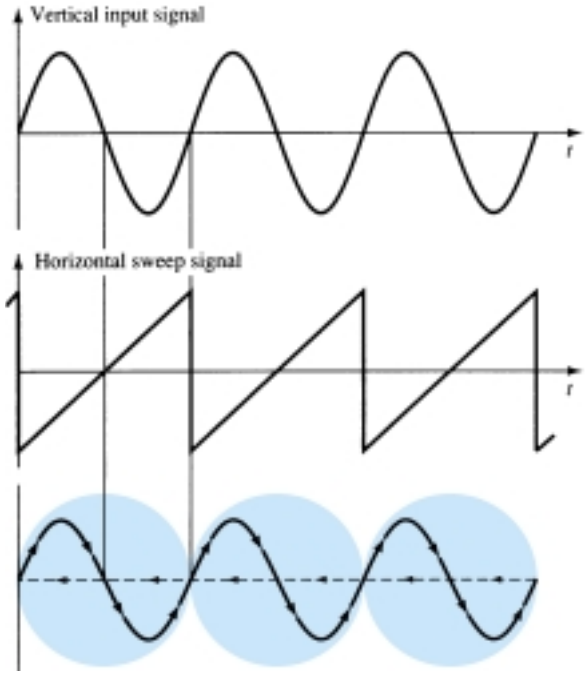


Figure 22.8 Steady scope display—input and sweep signals synchronized.

negative (retrace), the beam is blanked (the grid voltage prevents the electrons from hitting the tube face).

To see a steady display each time the beam is swept across the face of the tube, it is necessary to start the sweep at the same point in the input signal cycle. In Fig. 22.9, the sweep frequency is too low and the CRO display will have an apparent "drift" to the left. Figure 22.10 shows the result of setting the sweep frequency too high, with an apparent drift to the right.

It should be obvious that adjusting the sweep frequency to exactly the same as the signal frequency to obtain a steady sweep is impractical. A more practical proce-

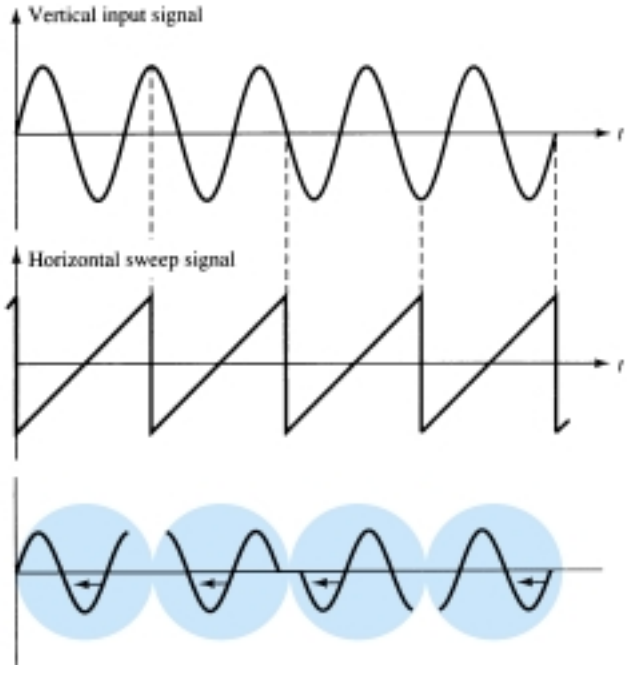


Figure 22.9 Sweep frequency too *low*—apparent drift to left.

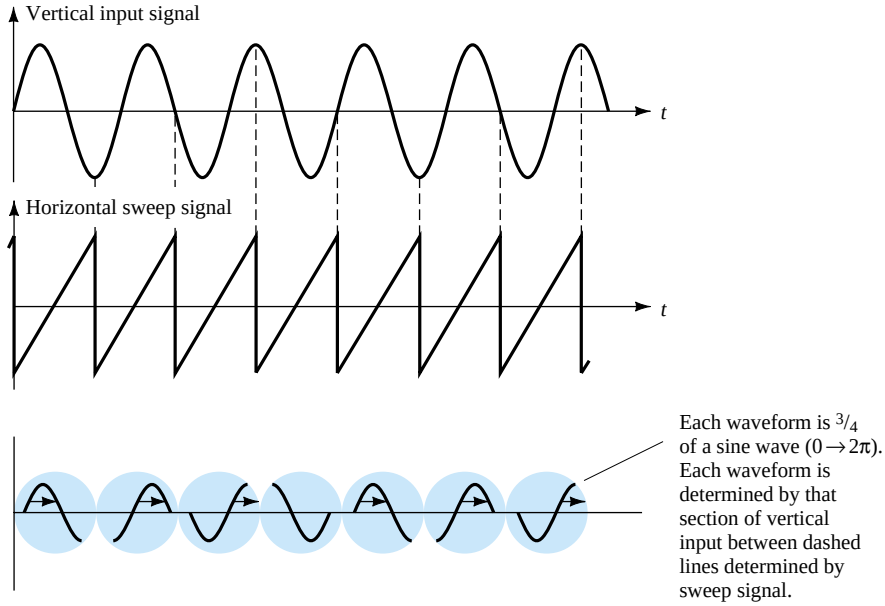


Figure 22.10 Sweep frequency too *high*—apparent drift to right.

dure is to wait until the signal reaches the same point in a cycle to start the trace. This triggering has a number of features, as described next.

Triggering

The usual method of synchronizing uses a portion of the input signal to trigger a sweep generator so that the sweep signal is locked or synchronized to the input signal. Using a portion of the same signal to be viewed to provide the synchronizing signal assures synchronization. Figure 22.11 shows a block diagram of how a trigger signal is derived in a single-channel display. The trigger signal source is obtained from the line frequency (60 Hz) for viewing signals related to the line voltage, from an external signal (one other than that to be viewed), or more likely, from a signal derived

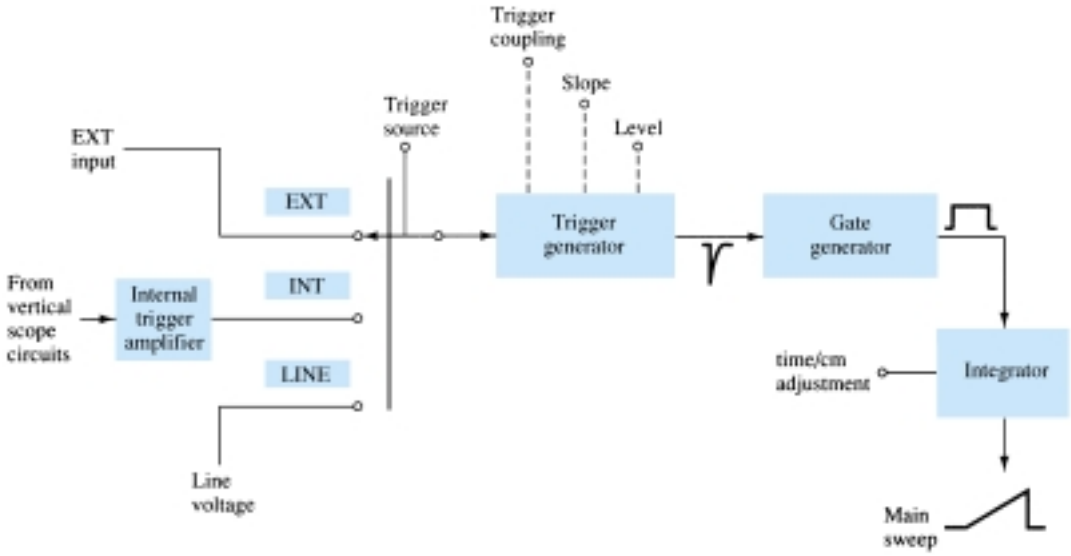


Figure 22.11 Block diagram showing trigger operation of scope.

from that applied as vertical input. The selector switch on the scope being set to INTERNAL will provide a part of the input signal to the trigger generator circuit. The output of the trigger generator is a trigger signal that is used to start the main sweep of the scope, which lasts a time set by the time/cm adjustment. Figure 22.12 shows triggering being started at various points in a signal cycle.

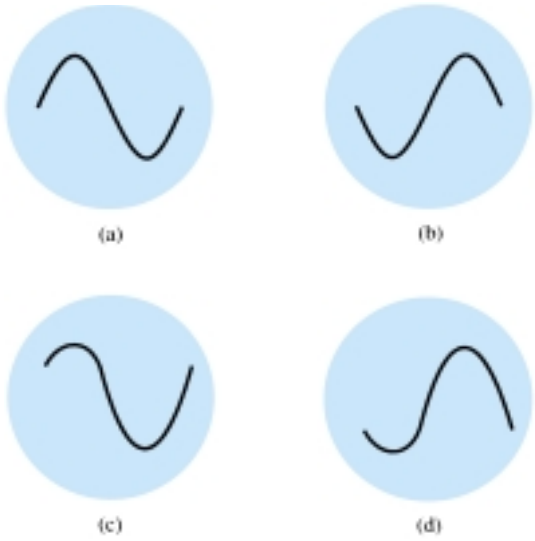


Figure 22.12 Triggering at various points of signal level (*Note:* sine starts at same point in cycle each sweep and is therefore synchronized): (a) positive-going zero level; (b) negative-going zero level; (c) positive-voltage trigger level; (d) negative-voltage trigger level.

The trigger sweep operation can also be seen by looking at some of the resulting waveforms. From a given input signal, a trigger waveform is obtained to provide for a sweep signal. As seen in Fig. 22.13, the sweep is started at a time in the input signal cycle and lasts a period set by the sweep length controls. Then the scope waits until the input reaches an identical point in its cycle before starting another sweep operation. The length of the sweep determines how many cycles will be viewed, while the triggering assures that synchronization takes place.

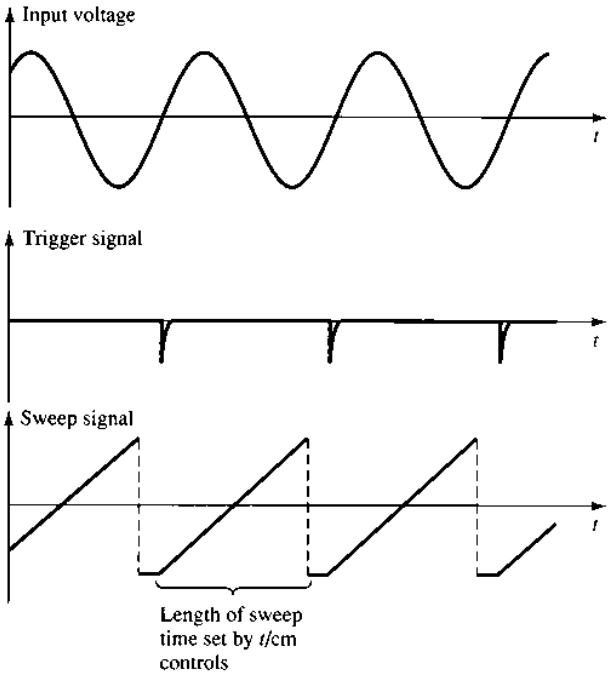


Figure 22.13 Triggered sweep.

22.6 MULTITRACE OPERATION

Most modern oscilloscopes provide for viewing two or more traces on the scope face at the same time. This allows comparing amplitude, special waveform features, and other important waveform characteristics. A multiple trace can be obtained using more than one electron gun, with the separate beams creating separate displays. More often, however, a single electron beam is used to create the multiple images.

Two methods of developing two traces are CHOPPED and ALTERNATE. With two input signals applied, an electronic switch first connects one input, then the other, to the deflection circuitry. In the ALTERNATE mode of operation, the beam is swept across the tube face displaying however many cycles of one input signal are to be displayed. Then the input switches (alternates) to the second input and displays the same number of cycles of the second signal. Figure 22.14a shows the operation with alternate display. In the CHOPPED mode of operation (Fig. 22.14b), the beam repeatedly switches between the two input signals during one sweep of the beam. As long as the signal is of relatively low frequency, the action of switching is not visible and two separate displays are seen.

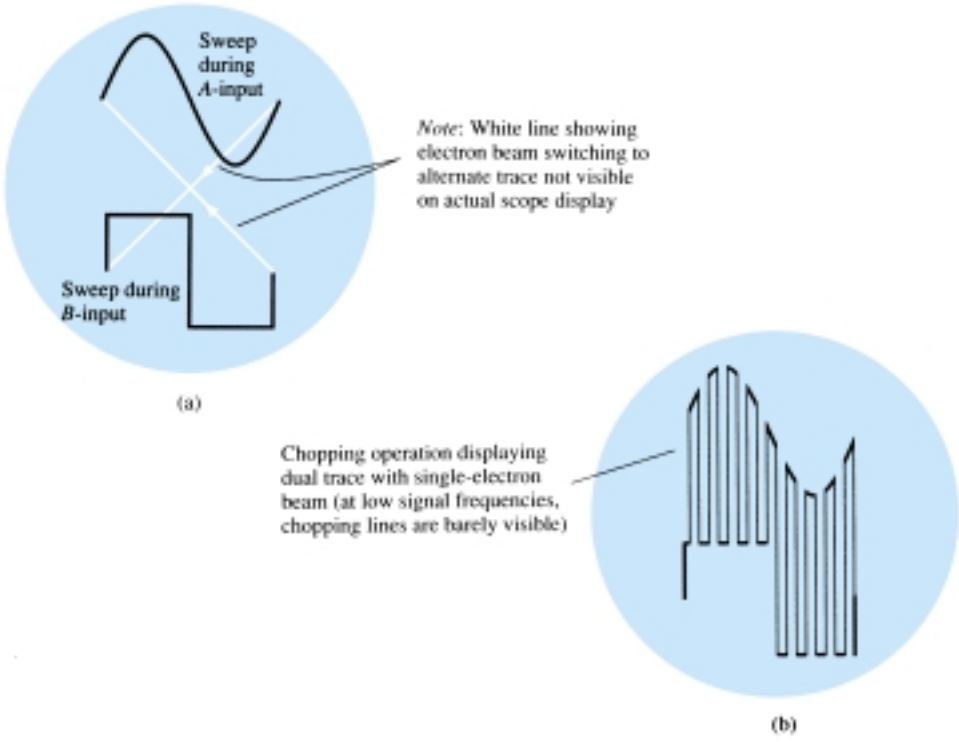


Figure 22.14 Alternate and chopped mode displays for dual-trace operation: (a) alternate mode for dual-trace using single electron beam; (b) chopped mode for dual-trace using single electron beam.

22.7 MEASUREMENT USING CALIBRATED CRO SCALES

The oscilloscope tube face has a calibrated scale to use in making amplitude or time measurements. Figure 22.15 shows a typical calibrated scale. The boxes are divided into centimeters (cm), 4 cm on each side of center. Each centimeter (box) is further divided into 0.2-cm intervals.

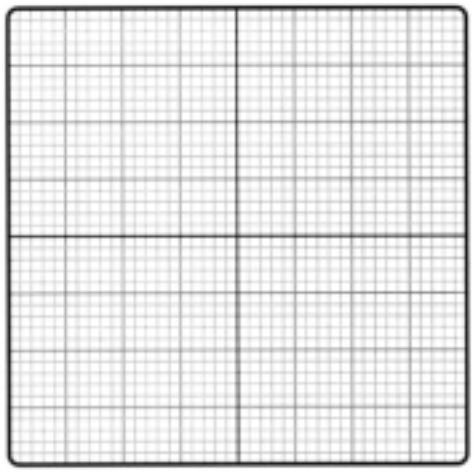


Figure 22.15 Calibrated scope face.

Amplitude Measurements

The vertical scale is calibrated in either volts per centimeter (V/cm) or millivolts per centimeter (mV/cm). Using the scale setting of the scope and the signal measured off the face of the scope, one typically can measure peak-to-peak or peak voltages for an ac signal.

EXAMPLE 22.2

Calculate the peak-to-peak amplitude of the sinusoidal signal in Fig. 22.16 if the scope scale is set to 5 mV/cm.

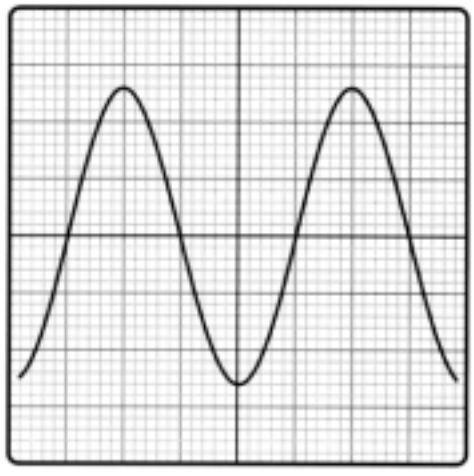


Figure 22.16 Waveform for Example 22.2.

Solution

The peak-to-peak amplitude is

$$2 \times 2.6 \text{ cm} \times 5 \text{ mV/cm} = \mathbf{26 \text{ mV}}$$

Note that a scope provides easy measurement of peak-to-peak values, whereas a multimeter typically provides measurement of rms (for a sinusoidal waveform).

Calculate the amplitude of the pulse signal in Fig. 22.17 (scope setting 100 mV/cm). *EXAMPLE 22.3*

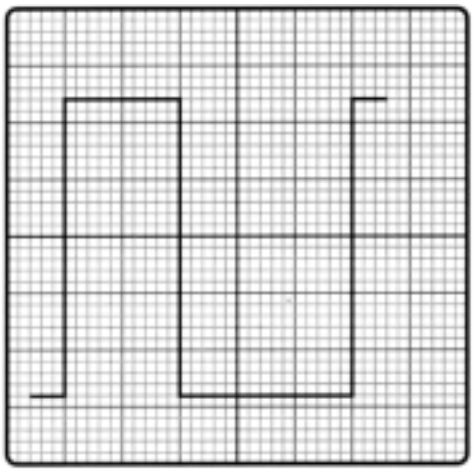


Figure 22.17 Waveform for Example 22.3.

Solution

The peak-to-peak amplitude is

$$(2.8 \text{ cm} + 2.4 \text{ cm}) \times 100 \text{ mV/cm} = \mathbf{520\ mV} = \mathbf{0.52\ V}$$

Time Measurements

PERIOD

The horizontal scale of the scope can be used to measure time, in seconds (s), milliseconds (ms), microseconds (μs), or nanoseconds (ns). The interval of a pulse from start to end is the period of the pulse. When the signal is repetitive, the period is one cycle of the waveform.

Calculate the period of the waveform shown in Fig. 22.18 (scope setting at 20 μs/cm). *EXAMPLE 22.4*

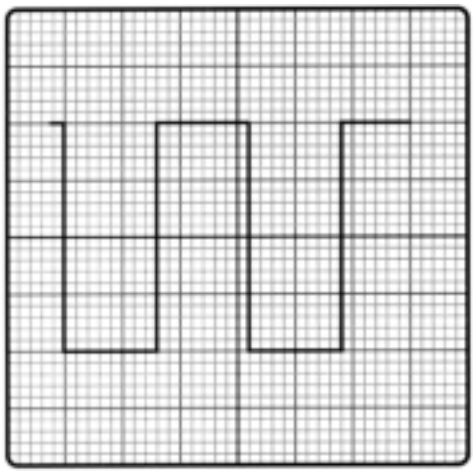


Figure 22.18 Waveform for Example 22.4.

Solution

For the waveform of Fig. 22.18,

$$\text{period} = T = 3.2 \text{ cm} \times 20\ \mu\text{s/cm} = \mathbf{64\ \mu s}$$

FREQUENCY

The measurement of a repetitive waveform's period can be used to calculate the signal's frequency. Since frequency is the reciprocal of the period,

$$f = \frac{1}{T} \tag{22.1}$$

EXAMPLE 22.5

Determine the frequency of the waveform shown in Fig. 22.18 (scope setting at 5 μs/cm).

Solution

From the waveform

$$\text{period} = T = 3.2 \text{ cm} \times 5\ \mu\text{s/cm} = 16\ \mu\text{s}$$

$$f = \frac{1}{T} = \frac{1}{16\ \mu\text{s}}$$

$$= \mathbf{62.5\ kHz}$$

PULSE WIDTH

The time interval that a waveform is high (or low) is the pulse width of the signal. When the waveform edges go up and down instantly, the width is measured from start (leading edge) to end (trailing edge) (see Fig. 22.19a). For a waveform with edges that rise or fall over some time, the pulse width is measured between the 50% points as shown in Fig. 22.19b.

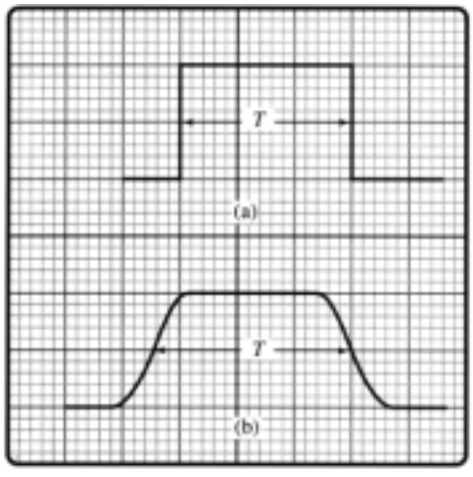


Figure 22.19 Pulse-width measurement.

Determine the pulse width of the waveform in Fig. 22.20. *EXAMPLE 22.6*

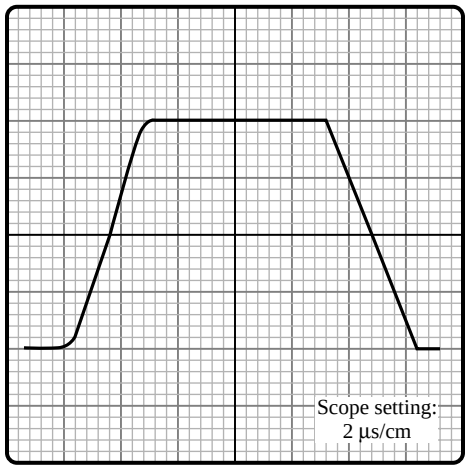


Figure 22.20 Waveform for Example 22.6.

Solution

For a reading of 4.6 cm at the midpoint of the waveform, the pulse width is

$$T_{PW} = 4.6 \text{ cm} \times 2 \ \mu\text{s/cm} = \mathbf{9.2\ \mu s}$$

PULSE DELAY

The time interval between pulses is called the pulse delay. For waveforms, as shown in Fig. 22.21, the pulse delay is measured between the midpoint (50% point) at the start of each pulse.

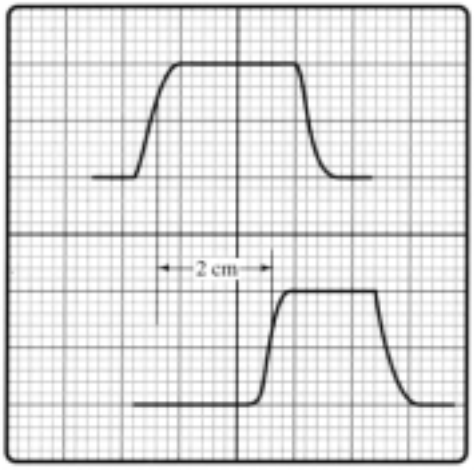


Figure 22.21 Waveform for Example 22.7.

Determine the pulse delay for the waveforms of Fig. 22.21. *EXAMPLE 22.7*

Solution

From the waveforms in Fig. 22.21,

$$\text{pulse delay} = T_{PD} = 2 \text{ cm} \times 50 \ \mu\text{s/cm} = \mathbf{100\ \mu s}$$

22.8 SPECIAL CRO FEATURES

The CRO has become more sophisticated and specialized in use. The range of amplitude measurements, the scales of time measurements, the number of traces displayed, the methods of providing sweep triggering, and the types of measurements are different depending on the area of specialized scope usage.

Delayed Sweep

A useful CRO feature uses two time bases to provide selection of a small part of the signal for viewing. One time base selects the overall signal viewed on the scope, while a second permits selecting a small part of the viewed signal to be displayed in an expanded mode. The main time base is referred to as the A time base, while the second time base, referred to as B, displays the signal after a selected delay time.

Figure 22.22 provides a block diagram showing the operation of the two time bases. With front-panel controls set to operate from the A sweep, a main sweep signal is set to view a number of cycles of the input signal. The controls then allow setting the B sweep using a variable setting dial, with the B sweep usually an intensified interval that can be moved over the face of the displayed sweep. When the desired portion of the displayed sweep is set, the controls are moved to display the delayed part of the signal, which is seen at the second time base setting as a magnified display. Figure 22.23 shows a pulse-type signal first viewed using the A sweep and then the selected portion on a magnified sweep setting.

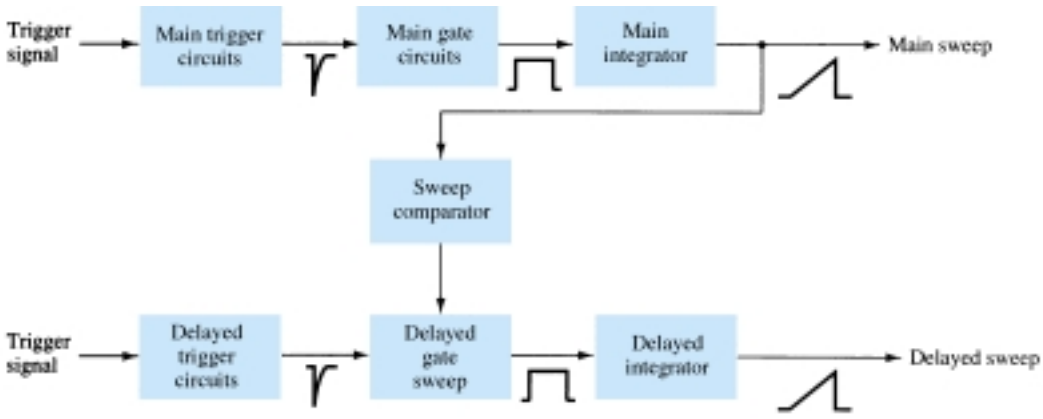


Figure 22.22 Operation of delayed sweep—block diagram.

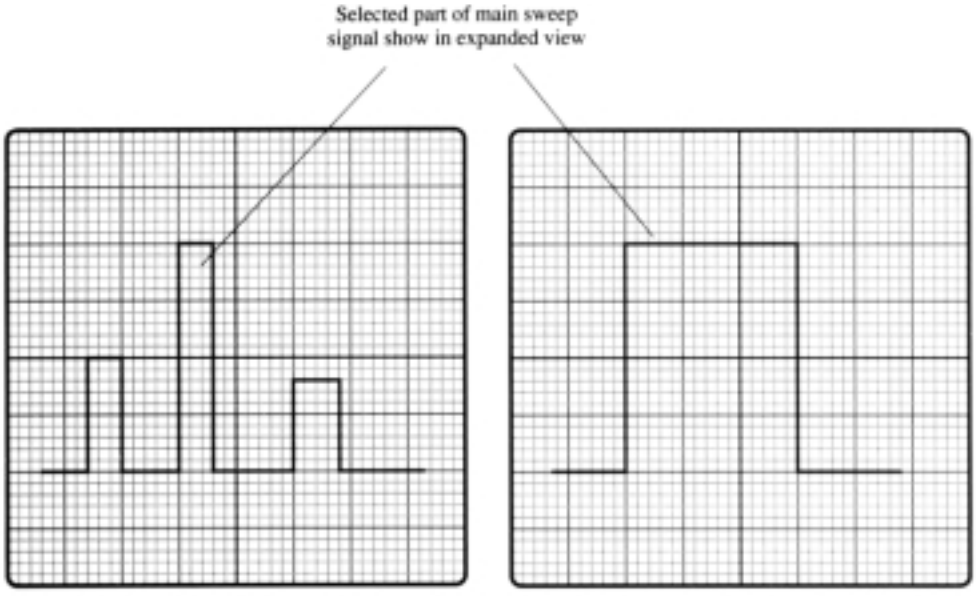


Figure 22.23 Main and delayed sweeps.

22.9 SIGNAL GENERATORS

A signal generator provides an ac signal of adjustable amplitude and varying frequency to use when operating an amplifier or other linear circuit. The frequency can typically be adjusted from a few hertz to a few megahertz. The signal amplitude can be adjusted from millivolts to a few volts of amplitude. While the signal is typically a sinusoidal waveform, pulse waveforms or even triangular waveforms are often available.

Waveform Generator IC(8038)

A precision waveform generator is provided by the 8038 IC unit shown in Fig. 22.24. The single 14-pin IC is capable of producing highly accurate sinusoidal, square, or triangular waveforms to use in operating or testing other equipment. Consideration of the IC's operation will help understand how any commercially available signal generator operates. This particular IC can provide output frequency that may be adjusted from less than 1 Hz up to about 300 kHz. The range of commercial units can be considerably higher. As indicated in Fig. 22.24, the IC provides three types of output waveform, and all at the same frequency, the frequency being selected by the user.

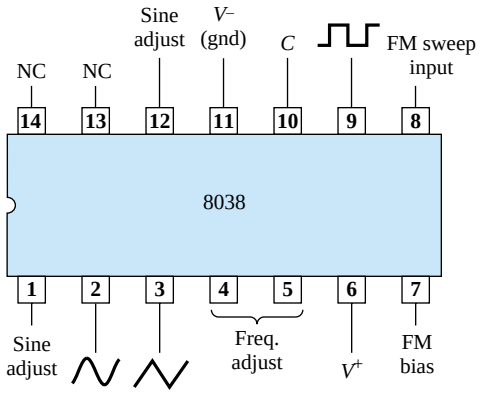


Figure 22.24 8038 waveform generator IC.

Figure 22.25 shows the connection of the IC when used to provide an adjustable frequency output. The frequency of the unit would then be

$$f = \frac{0.15}{RC} \tag{22.2}$$

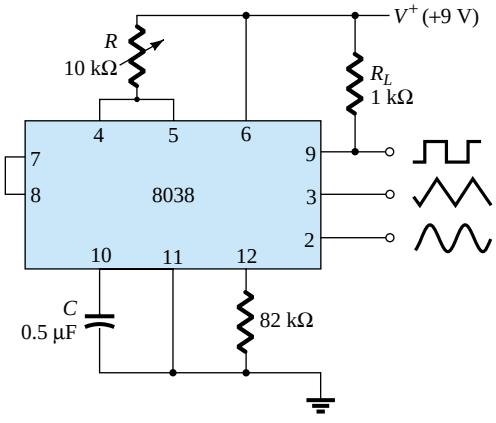


Figure 22.25 Connection of 8038 as variable frequency generator.

EXAMPLE 22.8

Referring to Fig. 22.25, determine the lowest and highest frequencies obtained when varying the 10-kΩ potentiometer from its minimum to its maximum setting.

Solution

Using Eq. (22.2), for a potentiometer set at 0, $R = 10\ \Omega$:

$$f = \frac{0.15}{(10\ \Omega)(0.5\ \mu\text{F})} = \mathbf{30\ kHz}$$

For a potentiometer set to its maximum,

$$f = \frac{0.15}{(10\ \text{k}\Omega)(0.5\ \mu\text{F})} = \mathbf{30\ Hz}$$

ADJUSTABLE OUTPUT AMPLITUDE

The connection of Fig. 22.26 shows how to provide adjustment of the sinusoidal waveform amplitude with the sinusoidal output provided through a buffered driver. The 310 op-amp is a unity-gain buffer providing the sinusoidal output from a low-impedance output. [The 310 has a voltage gain near unity (1), with an output impedance of about 1 Ω.] The output frequency is adjustable over a range from about 30 Hz to 30 kHz, with an amplitude adjustable up to about 9 V peak.

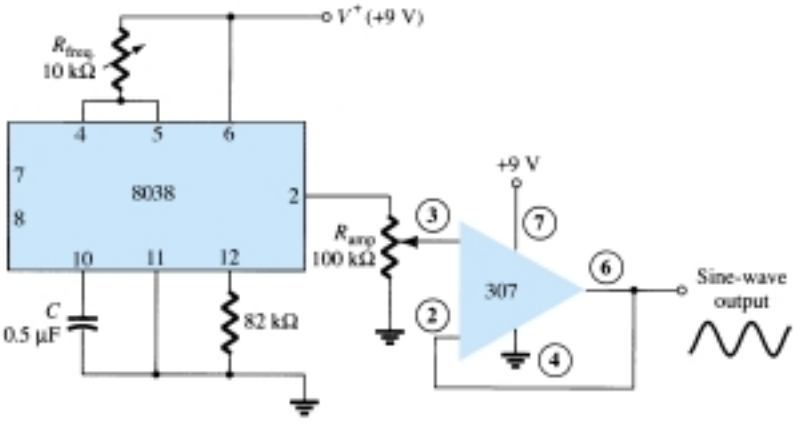


Figure 22.26 Sinusoidal waveform generator with adjustable frequency and amplitude.

5-V (TTL) PULSE GENERATOR

A circuit providing a 5-V pulse waveform for use with TTL digital circuits is shown in Fig. 22.27. The 8038 IC provides a rectangular or pulse waveform at a fixed output between 0 and +5 V. The frequency of the output can be varied from about 30 Hz to 30 kHz when adjusting the value of the 10-kΩ potentiometer. A commercial signal generator would probably include switched capacitors to provide frequency over a range of values. As long as the supply uses an IC regulator to provide the +5-V supply voltage, the output will be a well-defined value, as is typically used in TTL circuits. The 310 unity follower provides the output from a low-impedance source, making it possible to connect the output to a number of loads without affecting the amplitude or frequency of the signal waveform.

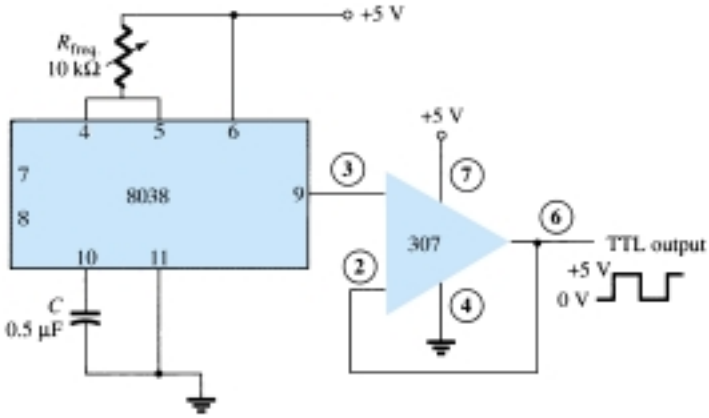


Figure 22.27 TTL signal waveform generator.